

# LDH-I Series

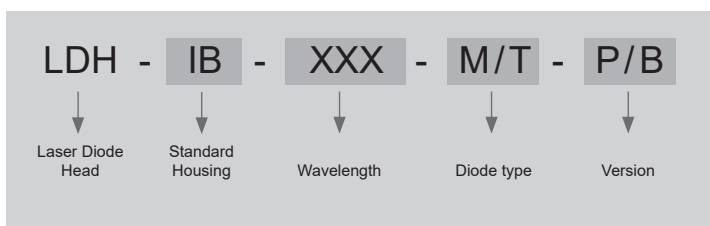
## Smart Laser Diode Heads for Taiko PDL M1

- **NEW** Wavelength 532, 561, and 594 nm
- **NEW** Design with improved efficiency
- **NEW** Dual calibration - max power and linear mode
- **NEW** High power multimode diodes in Vis and NIR
- Wavelengths between 375 and 1550 nm
- Pulse widths as short as 20 ps (FWHM)
- Repetition rate from single shot to 100 MHz
- Adjustable (average) power up to 200 mW
- Pulsed, bursts and CW operated
- Power calibrated, operating hours counter, TE cooled
- Collimated beam, optional fiber coupling



### Applications

- Time-resolved measurements
- Lifescience, FLIM, FCS, STED
- Materials science and semiconductor diagnostics
- Metrology and calibration
- Ranging and LiDAR
- Seeding



I = Taiko compatible, "intelligent"

M = Multimode diode, reduced coupling efficiency into single mode fibers, not suited for microscopy applications

T = Tapered amplified

B = High power

P = Narrow pulse

LDH-I Series features a large range of smart laser heads compatible with the Taiko PDL M1 driver. Laser heads from this series provide wavelengths in the spectral range from 375 to 1550 nm and include power calibrations. When coupled with a Taiko PDL M1, their optical output power can be controlled in both pulsed and continuous modes. Additionally, a wavelength calibration is available in CW mode. Each laser head is identified by the Taiko PDL M1 driver and includes not only linearized calibration data but also an operating hours counter.

The following tables list the pulse parameters and power values of the available wavelengths of the LDH-I Series (available between 375 and 1550 nm). The two power adjustment levels specified here refer to the same laser head. These levels can be adjusted using the corresponding driver Taiko PDL M1. The 'Low' adjustment is the best choice for shortest pulses and is usually reached close to the lasing threshold. The 'High' adjustment is used to achieve highest pulse power and corresponds to the maximum intensity setting of the driver. These laser heads have a spectral width of a few nm. Special selected laser heads with narrow spectral bandwidth can also be provided.

## Wavelengths

| Wavelength<br>(± 10) [nm] | Type<br>(LDH-)        | Pulse <sup>1</sup><br>(FWHM) [ps] | Max rep. rate<br>[MHz] | High avg. power <sup>2</sup><br>[mW] | Low avg. power <sup>3</sup><br>[mW] | CW power<br>[mW] |
|---------------------------|-----------------------|-----------------------------------|------------------------|--------------------------------------|-------------------------------------|------------------|
| 375                       | IB-375-P              | < 40                              | 70                     | 4.0                                  | 1.0                                 | 10               |
| 375                       | IB-375-B              | < 70                              | 80                     | 7.0                                  | 1.5                                 | 20               |
| 375                       | IB-375-M-P            |                                   |                        | On request                           |                                     |                  |
| 375                       | IB-375-M              |                                   |                        | On request                           |                                     |                  |
| 395                       | IB-390-B              | < 70                              | 100                    | 20.0                                 | 3.0                                 | 50               |
| 405                       | IB-405-P              | < 50                              | 80                     | 4.0                                  | 1.0                                 | 10               |
| 405                       | IB-405-B              | < 50                              | 100                    | 20.0                                 | 3.0                                 | 50               |
| 405                       | IB-405-M-P            |                                   |                        | On request                           |                                     |                  |
| 405                       | IB-405-M              |                                   |                        | On request                           |                                     |                  |
| 420                       | IB-420-B              | < 60                              | 100                    | 12.0                                 | 2.0                                 | 50               |
| 440                       | IB-440-B              | < 80                              | 100                    | 25.0                                 | 1.0                                 | 50               |
| 440                       | IB-440-M-P            |                                   |                        | On request                           |                                     |                  |
| 440                       | IB-440-M              |                                   |                        | On request                           |                                     |                  |
| 450                       | IB-450-B              | < 70                              | 80                     | 12.0                                 | 2.0                                 | 50               |
| 450                       | IB-450-M-P            | < 110                             | 50                     | 40.0                                 | 5.0                                 | 200              |
| 450                       | IB-450-M              | < 220                             | 50                     | 85.0                                 | -                                   | 200              |
| 470                       | IB-470-B              | < 70                              | 100                    | 15.0                                 | 1.0                                 | 70               |
| 470                       | IB-470-M-P            | < 110                             | 80                     | 40.0                                 | 15.0                                | 200              |
| 470                       | IB-470-M              | < 220                             | 50                     | 100.0                                | -                                   | 200              |
| 485                       | IB-485-P <sup>4</sup> | < 90                              | 60                     | 3.0                                  | 1.5                                 | 40               |
| 485                       | IB-485-B <sup>4</sup> | < 100                             | 100                    | 10.0                                 | 1.0                                 | 50               |
| 485                       | IB-485-M-P            | < 140                             | 50                     | 35.0                                 | 20.0                                | 200              |
| 485                       | IB-485-M              | < 220                             | 50                     | 100.0                                | -                                   | 200              |
| 500                       | IB-500-B              | < 110                             | 70                     | 7.0                                  | 1.8                                 | 40               |
| 510                       | IB-510-B <sup>4</sup> |                                   |                        | On request                           |                                     |                  |
| 515                       | IB-510-B <sup>4</sup> | < 160                             | 100                    | 7.0                                  | 1.0                                 | 20               |
| 520                       | IB-520-M-P            | < 130                             | 50                     | 13.0                                 | 6.0                                 | 200              |
| 520                       | IB-520-M              | < 200                             | 50                     | 50.0                                 | -                                   | 200              |
| 532 (± 3)                 | IB-530-T-P            | < 80                              | 100                    | -                                    | 0.5                                 | 20               |
| 532 (± 3)                 | IB-530-T-B            | < 80                              | 100                    | 1.1                                  | -                                   | 20               |
| 561 (± 3)                 | IB-560-T-P            | < 80                              | 100                    | -                                    | 0.5                                 | 20               |
| 561 (± 3)                 | IB-560-T-B            | < 80                              | 100                    | 0.9                                  | -                                   | 20               |
| 594 (± 3)                 | IB-595-T-P            | < 100                             | 100                    | -                                    | 0.2                                 | 5                |
| 594 (± 3)                 | IB-595-T-B            | < 100                             | 100                    | 0.5                                  | -                                   | 5                |
| 640                       | IB-640-B              | < 90                              | 100                    | 30.0                                 | 3.0                                 | 50               |
| 640                       | IB-640-M-P            | < 130                             | 60                     | 40.0                                 | 4.5                                 | 100              |
| 640                       | IB-640-M              | < 160                             | 70                     | 50.0                                 | -                                   | 100              |
| 655                       | IB-650-B              | < 70                              | 100                    | 10.0                                 | 2.5                                 | 10               |
| 670                       | IB-670-P              | < 50                              | 100                    | 2.5                                  | 1.0                                 | 5                |
| 670                       | IB-670-B              | < 70                              | 100                    | 7.0                                  | 0.8                                 | 10               |
| 685                       | IB-690-B              | < 80                              | 100                    | 10.0                                 | 1.0                                 | 20               |
| 705                       | IB-705-B              | < 70                              | 100                    | 20.0                                 | 3.0                                 | 25               |
| 730                       | IB-730-B              | < 80                              | 100                    | 20.0                                 | 3.0                                 | 20               |
| 780                       | IB-780-B              | < 70                              | 100                    | 30.0                                 | 1.0                                 | 40               |
| 780                       | IB-780-M-P            | < 150                             | 100                    | 40.0                                 | 12.0                                | 100              |
| 780                       | IB-780-M              | < 220                             | 80                     | 110.0                                | -                                   | 100              |
| 805                       | IB-810-B              | < 110                             | 80                     | 10.0                                 | 1.5                                 | 50               |
| 805                       | IB-810-M-P            | < 70                              | 100                    | 50.0                                 | 5.0                                 | 200              |
| 805                       | IB-810-M              | < 160                             | 80                     | 110.0                                | -                                   | 200              |
| 830                       | IB-830-B              | < 70                              | 100                    | 10.0                                 | 0.3                                 | 20               |
| 830                       | IB-830-M-P            | < 90                              | 100                    | 50.0                                 | 12.0                                | 200              |

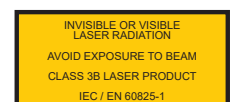
| Wavelength<br>(± 10) [nm] | Type<br>(LDH-) | Pulse <sup>1</sup><br>(FWHM) [ps] | Max rep. rate<br>[MHz] | High avg. power <sup>2</sup><br>[mW] | Low avg. power <sup>3</sup><br>[mW] | CW power<br>[mW] |
|---------------------------|----------------|-----------------------------------|------------------------|--------------------------------------|-------------------------------------|------------------|
| 830                       | IB-830-M       | < 160                             | 80                     | 110.0                                | -                                   | 200              |
| 850                       | IB-850-B       | < 90                              | 80                     | 10.0                                 | 3.0                                 | 50               |
| 905                       | IB-905-B       | < 100                             | 80                     | 8.0                                  | 2.0                                 | 50               |
| 910                       | IB-910-M-P     | < 100                             | 80                     | 30.0                                 | 10                                  | 200              |
| 910                       | IB-910-M       | < 180                             | 80                     | 150.0                                | -                                   | 200              |
| 940                       | IB-940-B       | < 90                              | 100                    | 6.0                                  | 2.0                                 | 50               |
| 975                       | IB-980-B       | < 100                             | 100                    | 7.0                                  | 1.8                                 | 50               |
| 980                       | IB-980-M-P     | < 80                              | 100                    | 35.0                                 | 4.0                                 | 200              |
| 980                       | IB-980-M       | < 180                             | 80                     | 100.0                                | -                                   | 200              |
| 1062                      | On request     |                                   |                        |                                      |                                     |                  |
| 1550 (± 10)               | IB-1550-B      | < 50                              | 100                    | 3.0                                  | 0.1                                 | 10               |

<sup>1</sup> Shortest pulse width at min intensity setting above laser threshold. Possible pulse broadening at high intensity settings. Pulses are deconvoluted with 30 ps detection IRF. Shorter pulse widths are available on demand.

<sup>2</sup> Average optical power at optimal repetition rate and max intensity setting (in max power mode).

<sup>3</sup> Average optical power at max repetition rate and min intensity setting above laser threshold (in linear and in max power mode).

<sup>4</sup> Different coupling efficiency into optical fibers for pulsed and CW operation due to astigmatism and possible wavelength shift. The coupling is optimized for pulsed operation as standard.



## Specifications

| Beam parameters                                |                                                                                   |
|------------------------------------------------|-----------------------------------------------------------------------------------|
| Optics focus length                            | $f' = 4.5 \text{ mm}$ (Typ. for most LDH-IB)                                      |
|                                                | $f' = 9.0 \text{ mm}$ (Typ. for LDH-IB-xxx-T)                                     |
| Numerical aperture                             | 0.55                                                                              |
| Typical divergence (with optics)               | theta parallel typ. 0.11 mrad<br>theta perpendicular typ. 0.32 mrad               |
| Polarisation                                   | typ. linear, perpendicular to the longer axis of the elliptical beam <sup>1</sup> |
| PER                                            | typ. $> 1:10$ ( $> 10 \text{ dB}$ )                                               |
| Side mode suppression ratio (SMSR)             | typ. $< 0.01$                                                                     |
| Cooling                                        |                                                                                   |
| Peltier cooling stability                      | better than 1 K for ambient temperature between 15°C and 30°C                     |
| Dimensions                                     |                                                                                   |
| Cylinder                                       | 76 × 175 mm (diameter × length)                                                   |
| Cylinder with fiber coupling                   | 76 × 207 mm (diameter × length)                                                   |
| Cuboid                                         | 175 × 77 × 83.7 mm (length × width × height)                                      |
| Cuboid with fiber coupling                     | 207 × 77 × 83.7 mm (length × width × height)                                      |
| Spectral width <sup>2</sup>                    |                                                                                   |
| Wavelengths $< 900 \text{ nm}$                 | approx. 2 to 8 nm                                                                 |
| Wavelengths $> 900 \text{ nm}$                 | approx. 10 to 20 nm                                                               |
| CW operation                                   | $< 1 \text{ nm}$                                                                  |
| Power stability (cooled)                       |                                                                                   |
| 12 hours, $\Delta T$ (ambient) $< 3 \text{ K}$ | 1 % RMS, 3 % peak to peak                                                         |

<sup>1</sup> A few exceptions may occur.

<sup>2</sup> Narrow bandwidth on request.



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